Newton's Laws of Motion

1. Inertia

Agree or Disagree?

1. "An object will eventually stop moving on its own".
2. "In order to keep something in motion, you have to keep applying force".

These statements may seem true, but we will see why they are not. Each of Newton's laws of motion addresses a common **misconception**.

NEWTON'S LAWS OF MOTION

1. INERTIA

"AN OBJECT IN MOTION WILL REMAIN IN MOTION, AND AN OBJECT AT REST WILL REMAIN AT REST, UNLESS ACTED UPON BY A FORCE."

What is inertia?

Inertia is an object's ability to resist any change in motion.

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Inertia is an object's ability to resist any change in motion.

Acceleration is a change in motion.

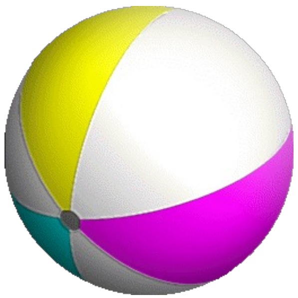
Inertia tells how much an object will resist acceleration.

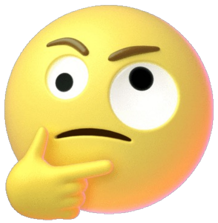
How can we measure inertia?

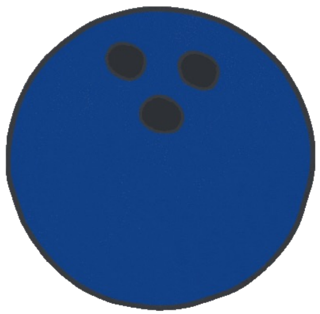
Inertia is an object’s ability to resist any change in motion.

Mass is the measure of inertia. More mass means more inertia.

Which ball would you rather kick?







How do objects start moving?

As long as the forces on an object are balanced, it will continue at the same velocity.

For the motion of an object to change, the forces must be **unbalanced.**

The unbalanced force must be strong enough and in the right direction to overcome an object's inertia.

How do objects stop moving?

On earth, there are a few forces that commonly stop objects

1. friction - whenever matter touches matter
2. air resistance - another kind of friction
3. Energy transformation - sound and thermal energy especially
4. gravity - objects are pulled toward the earth
5. the normal force - objects push back on each other

Are fast-moving objects harder to change?

Momentum is the product of an object's mass and velocity.

$$P = m \cdot v$$

Momentum is a vector quantity, with magnitude and direction.

Momentum is measured in units of kg $\frac{m}{s}$ (Newton-seconds)

Check Your Understanding

An object can start moving by itself.

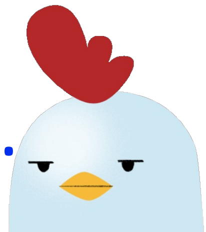
Inertia brings objects to a rest position.

Many different forces can act on an object at the same time.

An object’s inertia does not change with its speed.

Inertia is a force.

Without gravity, an object would have no inertia.



Explain why we wear seatbelts